

Solution to MP 352 Exam, Spring 2014

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P.1

This problem tests the understanding of not just Lorentz transformations, but also 4-vectors and the notation we use to describe them...

(a) We used this particular Lorentz so many times in lectures that it's something of a given question: if a boost is of speed $v = 0.6c$ in the x -direction, then $\gamma(0.6) = 1.25$ and so

$$\boxed{x'^0 = 1.25x^0 - 0.75x^1, x'^1 = 1.25x^1 - 0.75x^0, x'^2 = x^2, x'^3 = x^3} \quad [6 \text{ pts}]$$

(b) A 4-vector with upper (contravariant) indices has a norm-squared of

$$\|A\|^2 = -(A^0)^2 + (A^1)^2 + (A^2)^2 + (A^3)^2$$

so for $(1, -1, 0, 0)$, we have $\boxed{\|A\|^2 = 0}$, and thus that [3 pts]

vector is lightlike. A lower-index (covariant) 4-vector has [3 pts]

a norm-squared of

$$\|B\|^2 = -(B_0)^2 + (B_1)^2 + (B_2)^2 + (B_3)^2$$

so $(-1, 2, 0, 1)$ has $\boxed{\|B\|^2 = 4}$ and is thus spacelike. [3 pts each]

(c) 4-vectors with upper index transform exactly like (x^0, x^1, x^2, x^3) .

Thus, using the formula (a), we see that

$$\boxed{\begin{aligned} A'^0 &= 1.25(-1) - 0.75(1) = -2 \\ A'^1 &= 1.25(1) - 0.75(-1) = 2 \\ A'^2 &= 0 \\ A'^3 &= 0 \end{aligned}}$$

[10 pts]

Lower-index 4-vectors transform as if the sign of V were reversed, i.e.

$$\boxed{\begin{aligned} B'_0 &= \gamma(V) \left(B_0 + \frac{V}{c} B_1 \right) = 1.25(-1) + 0.75(2) = 0.25 \\ B'_1 &= \gamma(V) \left(B_1 + \frac{V}{c} B_0 \right) = 1.25(2) + 0.75(-1) = 1.75 \\ B'_2 &= B_2 = 0 \\ B'_3 &= B_3 = 1 \end{aligned}}$$

[10 pts]

(d) Now we check that the invariants $\|A\|^2$, $\|B\|^2$ and $A \cdot B = A^\mu B_\mu$ are indeed invariant. It's easy to see that

$$\boxed{\|A'\|^2 = -(2)^2 + (2)^2 = 0 = \|A\|^2}$$

[3 pts]

Also,

$$\|B'\|^2 = -(0.25)^2 + (1.75)^2 + 1 = 4 = \|B\|^2 \quad (2)$$

[3pts]

as expected. Fully,

$$A^M B_\mu = A^0 B_0 + A^1 B_1 + A^2 B_2 + A^3 B_3 = (1)(-1) + (-1)(2) + 0 + 0 = -3$$

and

$$A^M B'_\mu = (-2)(0.25) + (2)(1.75) + 0 + 0 = -0.5 + 3.5 = 3$$

So $A^M B'_\mu = A^M B_\mu$, as expected.

[6pts]

P.2

Although this question looks a bit out of place, I used some of this course to introduce the concept of groups to the students, both in a general way and in the context of this course (e.g. $SO(3,1)$, the Lorentz group). Since this is the only place some students might see group theory before getting their degrees, I figured it was worth including.

(a) We went through these with several examples, so hopefully most students will remember most of the basic properties:

- | | |
|--|--------|
| 1. CLOSURE: if a and b are in G , then $a * b$ is also in G | [5pts] |
| 2. ASSOCIATIVITY: for any $a, b, c \in G$, $(a * b) * c = a * (b * c)$ | [5pts] |
| 3. IDENTITY: there must exist an element e in G such that, for all $a \in G$, $a * e = e * a = a$. | [5pts] |
| 4. INVERSE: for each $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$ | [5pts] |

(b) In lecture, we showed that $\mathbb{R} \setminus \{0\}$ was a group under multiplication and $\mathbb{Z}$ a group under addition; this just abstracts that a bit by putting the ticks.

1. CLOSED? If x, y are both positive, the so is xy . If m and n are both integers, so is $m+n$. Thus, $(xy, m+n) \in G$, so $(G, +)$ satisfies closure.

[6pts]

2. ASSOCIATIVE? Multiplication and addition are both associative, so

$$\begin{aligned} ((x, k) * (y, m)) * (z, n) &= (xy, k+m) * (z, n) \\ &= (xyz, k+m+n) = (x, k) * (yz, m+n) = (x, k) * ((y, m) + (z, n)) \end{aligned}$$

and so $*$ is indeed associative.

[8pts]

(3)

3. IDENTITY? We want to find (e_1, e_2) which satisfies
 $(x, m) * (e_1, e_2) = (e_1, e_2) * (x, m) = (x, m)$. We see that
 using $(xe_1, m+e_2) = (x, m)$, so $e_1 = 1$ and $e_2 = 0$. Thus,

$(1, 0)$ is the identity element.

[8 pts]

4. INVERSE? $(x, m)^{-1}$ is some pair (y, n) satisfying

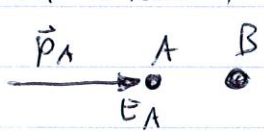
$$(x, m) * (y, n)^{-1} = (x, m)^{-1} * (x, m) = e = (1, 0)$$

or $xy = 1, m+n = 0$. Thus, $y = 1/x, n = -m$ but we
 also have to state that because $x > 0$, $1/x$ exists and is positive,
 and since $m \in \mathbb{Z}$, $-m$ is also an integer. Thus $(1/x, -m)$ is
 both an inverse and in G , so $(x, m)^{-1} \in G$ does exist. [8 pts]

P.3

In this problem, we test concepts of "rest frame", "rate-of-mass frame"
 as well as invariance of $\|P\|^2$ of warblers' particles.

(a) If $A \sim B$ has a nonzero mass, then we can go into its
 rest frame. Suppose without loss of generality we assume $m_B > 0$.
 Then the total 4-momentum in the rest frame of B is



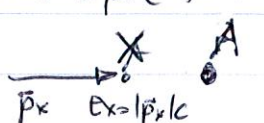
$$P_m^M = \begin{pmatrix} E_A/c + m_B c \\ \vec{p}_A \end{pmatrix}$$

where $E_A = \sqrt{|\vec{p}_A|^2 c^2 + m_A^2 c^4}$. Thus, $\|P\|^2 = -(E_A/c + m_B c)^2 + |\vec{p}_A|^2$

$$= -\frac{E_A^2}{c^2} - 2m_B E_A - m_B^2 c^2 + |\vec{p}_A|^2 = -(2m_B E_A + m_A^2 c^2 + m_B^2 c^2)$$

But if X is warbler, then $\|P_{\text{tot}}\|^2 = 0$. Thus, if this event can occur,
 $2m_B E_A + m_A^2 c^2 + m_B^2 c^2 = 0$. But $m_B > 0$, and since $m_A \geq 0$ and $E_A \geq 0$,
 we see there is no way this can be satisfied. Thus, if either particle [8 pts]
is warbler, this reaction cannot occur.

(b) Now we assume $m_B > m_A > 0$. Since $m_A > 0$, A has a rest frame, so
 if we could go to 4-momentum in the rest frame of A , we see



$$P_m^M = \begin{pmatrix} |\vec{p}_X|c + m_A c \\ \vec{p}_X \end{pmatrix}$$

$$\Rightarrow \|P_m\|^2 = -(|\vec{p}_X|c + m_A c)^2 + |\vec{p}_X|^2 = -2|\vec{p}_X|m_A c - m_A^2 c^2.$$

Now, $\|P_m\|^2 = \|P_{\text{tot}}\|^2$, and we know P_{tot} is always future-like: if we use B 's
 rest frame, then $P_{\text{tot}}^M = \begin{pmatrix} m_B c \\ 0 \end{pmatrix}$, so $\|P_{\text{tot}}\|^2 = -m_B^2 c^2$. Thus, $\|P_m\|^2 = \|P_{\text{tot}}\|^2$

(4)

gives $|\vec{p}_x| = \frac{m_B^2 - m_A^2}{2m_A} c$. (Note that if $m_B \leq m_A$, $|\vec{p}_x| \leq 0$, which is not possible, and so $m_B > m_A$ is necessary). Now, if A is rest frame, the total 3-mom goes in $\vec{p}_x$, so $\vec{p}_x$ is also the total 3-mom goes at, i.e. B's mom. Thus, the energy of B is then given by

$$E_B = \sqrt{|\vec{p}_x|^2 c^2 + m_B^2 c^4} = \sqrt{\left(\frac{m_B^2 - m_A^2}{2m_A}\right)^2 c^4 + m_B^2 c^4}$$

$$= \sqrt{\frac{m_B^4 - 2m_B^2 m_A^2 + m_A^4 + 4m_A^2 m_B^2}{4m_A^2}} c^2$$

$$= \sqrt{\left(\frac{m_A^2 + m_B^2}{2m_A}\right)^2 c^2} = \boxed{\frac{m_A^2 + m_B^2}{2m_A} c^2} \quad [70pts]$$

(c) Here, we're told that the total CM energy is E_{CM} . Find the energies of the two X's in CM: it's a CM frame, these will have equal and opposite momenta, $\vec{p}_x$ and $-\vec{p}_x$. Since their energies are both $|\vec{p}_x|c$, then we

$$P_{out, CM} = \begin{pmatrix} 2|\vec{p}_x| \\ 0 \end{pmatrix}. \text{ But } \|P_{out, CM}\|^2 = -E_{CM}^2/c^2 = -4|\vec{p}_x|^2 c^2,$$

$$\text{so } |\vec{p}_x| = \frac{E_{CM}}{2c}, \text{ or } \boxed{E_x = \frac{1}{2} E_{CM}}. \quad [70pts]$$

A & B have equal and opposite momenta in the CM frame, so

$$E_A = \sqrt{|\vec{p}_A|^2 c^2 + m_A^2 c^4} \text{ and } E_B = \sqrt{|\vec{p}_A|^2 c^2 + m_B^2 c^4}. \text{ Thus,}$$

$$\sqrt{|\vec{p}_A|^2 c^2 + m_A^2 c^4} + \sqrt{|\vec{p}_A|^2 c^2 + m_B^2 c^4} = E_{CM}. \text{ But,}$$

$$(\sqrt{|\vec{p}_A|^2 c^2 + m_A^2 c^4})^2 = (E_{CM} - \sqrt{|\vec{p}_A|^2 c^2 + m_B^2 c^4})^2$$

or

$$|\vec{p}_A|^2 c^2 + m_A^2 c^4 = E_{CM}^2 - 2E_{CM} \sqrt{|\vec{p}_A|^2 c^2 + m_B^2 c^4} + |\vec{p}_A|^2 c^2 + m_B^2 c^4.$$

Thus we

$$m_A^2 c^4 = E_{CM}^2 - 2E_{CM} E_B + m_B^2 c^4$$

$$\text{so } \boxed{E_B = \frac{E_{CM}^2 + m_B^2 c^4 - m_A^2 c^4}{2E_{CM}}} \quad [70pts]$$

Finally,

$$E_A = E_{CM} - E_B = \frac{2E_{CM}^2 - (E_{CM}^2 + m_B^2 c^4 - m_A^2 c^4)}{2E_{CM}}$$

$$= \boxed{\frac{E_{CM}^2 + m_A^2 c^4 - m_B^2 c^4}{2E_{CM}}} \quad [70pts]$$